LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

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We give a negative answer to the longstanding question whether every countable group is MF. Here MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. Our main tool is the following one-sided compression criterion. Let G be countable, let $L \leq G$ have property (T), and let $K \trianglelefteq G$ have property (T). If K is contained in the normal subgroup generated by the commutators $[ucu^{-1}, \ell]$, where $uLu^{-1} \leq L$, $c \in C_G(L)$, and $\ell \in L$, then every homomorphism from G to an MF group kills K. For $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$, a single commutator of this form normally generates H. The group H is nontrivial, simple, and has property (T), whereas every homomorphism from H to an MF group is trivial. In particular, H is not MF. Consequently, the reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF. Finally, if $f\colon G \to Q$ is surjective and $\ker f \leq \mathrm{Rad}_{\mathrm{MF}}(G)$, then $\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q))$. In this case, image and inverse image identify the normal subgroups $N \trianglelefteq G$ satisfying $\ker f \leq N$ and G/N MF with the normal subgroups $P \trianglelefteq Q$ satisfying Q/P MF. Finally, the MF property of finitely presented groups is Π^0_2-complete to recognize: no algorithm decides it from a finite presentation, and the finite presentations of non-MF groups cannot be recursively enumerated. The lower bound compiles the halting hierarchy into finite presentations by HNN extensions, kept MF by a permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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1. Introduction

Not every countable group is MF. Our counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)).$$

The group H is countable, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is

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not MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

Thus the models become multiplicative in operator norm without losing any nonidentity element. This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum.

A separable C^*-algebra is called MF if it embeds as a C^*-subalgebra of a norm matrix corona. For every countable group G, the reduced group algebra $C^*_{\mathrm{r}}(G)$ is separable and has a faithful canonical trace, hence is stably finite. Suppose that $e\colon C^*_{\mathrm{r}}(G) \to \mathcal{Q}_{\mathbf{d}}$ is an MF embedding and put $p = e(1)$. Then $g \mapsto e(\lambda_g) + (1-p)$ is an injective homomorphism from G into $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, so G is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group C^*-algebra that is not MF.

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \ \bigcap_{\pi\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

Proposition 1.1 shows that G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF kernel closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap \{\ker f : N \le \ker f,\ f\colon G \to M,\ M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 1.1 (basic properties of the MF radical)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \quad \Longleftrightarrow \quad \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and π is a corona homomorphism of G, then $\pi \circ \alpha$ kills x. Hence $\pi(\alpha(x)) = 1$ for every π, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$. If G/R is trivial, there is nothing to prove. If $x \in G/R$ is nontrivial and $g \in G$ maps to x, then $g \notin R$. Some homomorphism from G to an MF group therefore separates g from the identity. It kills R and descends to a homomorphism of G/R that separates x. Thus G/R is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

For $N \trianglelefteq G$, homomorphisms from G/N to MF groups correspond to homomorphisms from G to MF groups that kill N. It follows directly that

$$\mathrm{Rad}_{\mathrm{MF}}(G/N) = \mathrm{cl}^G_{\mathrm{MF}}(N)/N.$$

Since a countable group is MF exactly when its MF radical is trivial, this identity gives the asserted closure criterion. □

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) *and $\mathfrak{D}_G(L) = G$, then* $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

We sketch the argument. Corollary 3.4 gives $\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G)$. Equivalently, if (V_n) is any operator norm asymptotic representation of G, then $\|V_n(d) - 1\|_2 \to 0$ for every $d \in \mathfrak{D}_G(L)$. To upgrade this conclusion to corona triviality on K, fix a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ and suppose, toward a contradiction, that Θ is nontrivial on K. Let p be the image in $\mathcal{Q}_{\mathbf{d}}$ of the Kazhdan projection of K. Because $K \trianglelefteq G$, the projection p commutes with $\Theta(G)$; hence $q = 1 - p$ defines a G-invariant complementary corner. After restricting coordinates and identifying each nonzero corner with a full matrix algebra, Lemma 4.1 realizes this action by an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$. Its restriction to K has no nonzero fixed vectors, so the Kazhdan inequality yields a single $s_0 \in K$ and a subsequence along which $\|W_n(s_0) - I_{r_n}\|_2$ stays bounded below by a positive constant. This contradicts $s_0 \in K \leq \mathfrak{D}_G(L)$ and the preceding convergence. Thus every corona homomorphism kills K, as required.

Theorem 2.1 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

We next apply Theorem A to the binary Leavitt construction that produces the group H announced above.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is countable, nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \mathrm{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1t_1)s_1 = 1$ shows that $d \neq 1$. Since H is simple, d normally generates H. Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)^{-1}$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 3.3.

A full MF radical also lets us prescribe the quotient visible to MF.

Theorem C (prescribed quotients visible to MF)**.** *Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$ and, for a countable group Q, define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q\colon W_Q \to Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

The final main result concerns all finitely presented groups at once. It locates the recognition problem for the MF property in the Kleene–Mostowski arithmetical hierarchy, whose levels Σ^0_n and Π^0_n count alternations of quantifiers over a decidable relation.

Theorem D (recognizing MF groups)**.** *Fix an effective coding of finite presentations of groups by natural numbers. The set of codes of finite presentations of MF groups is Π^0_2-complete, and the set of codes of finite presentations of non-MF groups is Σ^0_2-complete. In particular, neither set is recursively enumerable, and no algorithm decides from a finite presentation whether the presented group is MF. More precisely, there is a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the group presented by $\widehat{R}_e$ is MF if and only if the e-th partial computable function has infinite domain.*

The upper bound is a certificate normal form: the MF property is determined by finite data with a fixed separation constant, and the existence of a certificate at a given scale is decidable (Section 8). The lower bound follows the architecture of Higman's embedding theorem. A switch turns e into a recursively presented group that is trivial or contains H according as the domain is infinite or finite; two HNN extensions wrap it into a finitely presented group (Section 10); and the extensions stay MF on the positive branch because of a permanence theorem, Theorem 9.2: an HNN extension of a group with a regular trace remains MF whenever the edge isomorphism is implemented by a unitary of a norm matrix corona (Sections 9 and 11).

Relation to prior work. Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite C^*-algebras might coincide. Their 1997 paper developed these notions and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite C^*-algebra is quasidiagonal. This was a question about nuclear C^*-algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite C^*-algebra were MF, then every countable group would be MF because its reduced group C^*-algebra is separable and stably finite. Later approximation papers described the resulting operator norm group problem as related to Kirchberg's question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite C^*-algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group C^*-algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that $A *_C A$ is MF whenever A is a separable MF C^*-algebra and $C \subseteq A$ is a C^*-subalgebra [Sh, Theorem 10]. She also proves that the full group C^*-algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten p-norms with $1 < p < \infty$ [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for $p = 1$ [BDL]. The operator norm case $p = \infty$ remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]. Here operator norm control implies that coordinatewise conjugation defines a representation in a second norm matrix corona. The Kazhdan projection of this conjugation representation makes the Hilbert–Schmidt asymptotic commutant invariant under the one-sided compressor. Theorem 4.2 then proves that a normal property-(T) subgroup contained in the defect lies in the MF radical.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of GL_n over an exchange ring that are normalized by EL_n [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality $Z(R) = \mathbb{F}_2$, implies that $\mathrm{EL}_{12}(R)$ is simple (Proposition 6.1). Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov, and Adian–Rabin gives undecidability; Theorem D gives the exact level. Torsion-freeness is the only other property of finitely presented groups whose recognition problem was known to be Π^0_2-complete, by Lempp [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is Π^0_2-hard to recognize [BCR]; their construction is the switch of Section 10. The passage from recursive to finite presentations is where the operator algebra enters: a generic Higman embedding does not preserve the MF property, and the ropes of Section 10 are designed so that Theorem 9.2 applies to them.

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \le G$, and suppose $u \in G$ satisfies $uLu^{-1} \le L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. □

This is where finite dimensionality enters the proof: an injective linear self-map of an infinite-dimensional commutant need not be surjective. In Section 3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity for projections onto fixed vectors.

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator norm asymptotic representations of G.

For $L \leq G$, let

$$\mathcal{C}_2(V,L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 3.1. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^*x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^*-algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1-q)$ is an isometry in A. Consequently $ww^* = q$, and hence $p = q$. □

Lemma 3.2 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\mathrm{Fix}\,\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\,\mathrm{Fix}\,\pi(L) \subseteq \mathrm{Fix}\,\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. □

Theorem 3.3 (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\mathrm{Ad}(V(u))(\mathcal{C}_2(V,L)) = \mathcal{C}_2(V,L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|\,,$$

so the operator norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\mathrm{Fix}\,\sigma(L)$. This representation of $\mathcal{B}$ need not be faithful; it is used only after the following projection identity has been proved inside $\mathcal{B}$. Since $uLu^{-1} \le L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$, so U commutes with P. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\mathrm{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. □

Corollary 3.4. *Let $L \le G$ have property* (T)*, let $uLu^{-1} \le L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 3.3, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\left\| V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1}) \right\| \longrightarrow 0.$$

Therefore

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

4. Normal Kazhdan subgroups and the MF radical

Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona representation of G. Normality of K implies that p commutes with the image of G. Hence $q = 1 - p$ also commutes with that image, and in every nondegenerate representation of the corner $q\mathcal{Q}_{\mathbf{d}}q$ the induced representation of K has no nonzero fixed vectors.

Lemma 4.1 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \mathrm{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_nW_n(g)J_n^*)$ is the coordinate restriction of $q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_nU_n(g) - U_n(g)q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_nU_n(g)q_n$, viewed on $q_n\mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g, let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\left\|\widetilde{W}_n(g) - q_nU_n(g)q_n\right\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel those coordinates. Put $r_n = \mathrm{rank}(q_n)$, choose a unitary $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^*\widetilde{W}_n(g)J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_nW_n(g)J_n^*)$ with the coordinate restriction of $q\rho(g)$. □

Theorem 4.2 (normal Kazhdan radical theorem)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T) *with $K \leq R_{\infty\to 2}(G)$. Then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K-fixed vectors. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g)\operatorname{Fix}\Theta(K)$ and $\operatorname{Fix}\Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K, so every element of K has trivial corona image. Lemma 4.1 gives positive integers r_n and an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$. Let $\widehat{\Theta}\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. The coordinatewise corner identifications carry $\widehat{\Theta}$ to the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. The Kazhdan projection of K under $\widehat{\Theta}$ is zero. Hence the positive element

$$b = \frac{1}{|S|}\sum_{s\in S}(\widehat{\Theta}(s) - 1)^*(\widehat{\Theta}(s) - 1)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|}\,1.$$

Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. Because the image of the Kazhdan projection is zero, $\widehat{\Theta}|_K$ has no nonzero fixed vectors. The defining Kazhdan inequality therefore gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s) - I_{r_n})^*(W_n(s) - I_{r_n})$$

represent b. Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from I_{r_n}. This contradicts $s_0 \in K \leq R_{\infty\to 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K, as required. $\square$

Proof of Theorem A. For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 3.4 gives

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$

Normality of $R_{\infty\to 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \le R_{\infty\to 2}(G).$$

Since $K \le \mathfrak{D}_G(L)$, Theorem 4.2 now gives $K \le \mathrm{Rad}_{\mathrm{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator of $\mathfrak{D}_G(L)$. Its kernel is normal, so it contains $\mathfrak{D}_G(L)$. ☐

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{4}$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{5}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0 q = 0$, and $t_0 s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0 I_3 t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0 \Psi(A) s_0 = A$, and hence Ψ is injective. Thus Ψ is an injective identity-preserving multiplicative map. It is not additive, since $\Psi(0) = qI_3 \neq 0$. Consequently, it restricts to an injective group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$.

To implement Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{6}$$

A direct block multiplication gives

$$\tau\ \mathrm{diag}(A, I_9)\, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{7}$$

Lemma 5.1. *The matrix τ defined in (6) belongs to* $\mathrm{EL}_{12}(R)$.

Proof. Put $I = I_6$. The Whitehead factorization gives

(8)
$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

If $B = (b_{rs})_{r,s=0}^{5} \in M_6(R)$, then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{r,6+s}(b_{rs}), \qquad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (8) therefore belongs to $\mathrm{EL}_{12}(R)$. Since $Y = X^{-1}$, its left-hand side is τ, and the result follows. □

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \le H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 5.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \le L. \tag{9}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 5.1 gives $\tau \in H$.

For $i \ne j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L. These matrices generate L, so (9) follows. □

6. Simplicity and the full MF radical

Proposition 6.1. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \mathrm{EL}_{12}(R)$, the subgroup $N \le \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R, I) \le N \le C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\mathrm{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \le C_{12}(R, 0) = Z(\mathrm{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix λI_{12}. Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^{\times}$. Since

$Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. ☐

Proposition 6.2. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L, so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{10}$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Its normal closure is a nontrivial normal subgroup of H, and hence is all of H by Proposition 6.1. ☐

Proof of Theorem B. The finitely generated $\mathbb{F}_2$-algebra R is countable, and hence so is H. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (9), the centrality of c relative to L, and Proposition 6.2 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 6.2 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Independently of this radical calculation, Proposition 6.1 shows that H is nontrivial and simple.

If M is MF and $f \colon H \to M$ is a homomorphism, compose f with a faithful corona embedding of M. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C^*_{\mathrm{r}}(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding $e \colon C^*_{\mathrm{r}}(H) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1 - p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. This contradicts the fact that H is not MF. ☐

7. Quotients visible to MF

The MF radical is exact across every surjection whose kernel is already invisible to MF.

Proposition 7.1 (full-kernel pullback)**.** *Let $f\colon G \to Q$ be a surjective homomorphism of countable groups. If* $\ker f \le \operatorname{Rad}_{\mathrm{MF}}(G)$*, then*

$$\operatorname{Rad}_{\mathrm{MF}}(G) = f^{-1}(\operatorname{Rad}_{\mathrm{MF}}(Q)).$$

In particular, the conclusion holds whenever $\operatorname{Rad}_{\mathrm{MF}}(\ker f) = \ker f$*. More generally, for every normal subgroup* $N \trianglelefteq G$*,*

$$\operatorname{cl}^G_{\mathrm{MF}}(N) = f^{-1}\Big(\operatorname{cl}^Q_{\mathrm{MF}}(f(N))\Big)\,.$$

Consequently,

$$G/N \text{ is MF} \quad\Longleftrightarrow\quad \ker f \le N \ \text{ and } \ Q/f(N) \text{ is MF.}$$

Proof. Every corona homomorphism from G kills $\ker f$ and therefore factors uniquely through f. Conversely, composing a corona homomorphism of Q with f gives one of G. Intersecting these two corresponding families of kernels gives the displayed MF radical pullback. For the stated sufficient condition, let $\pi\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be any corona homomorphism. Its restriction to $\ker f$ is trivial because $\operatorname{Rad}_{\mathrm{MF}}(\ker f) = \ker f$. Hence $\ker f \le \operatorname{Rad}_{\mathrm{MF}}(G)$.

The same factorization, restricted to homomorphisms that kill N, gives the closure formula. If this closure equals N, then it contains $\ker f$. Mapping the equality onto Q shows that $\operatorname{cl}^Q_{\mathrm{MF}}(f(N)) = f(N)$. Conversely, these two conditions give

$$\operatorname{cl}^G_{\mathrm{MF}}(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 1.1, applied in G and Q. □

Thus image under f and inverse image under f give mutually inverse correspondences, preserving inclusion between the normal subgroups with MF quotient in G and in Q. The map induced by f on the largest quotients visible to MF is an isomorphism:

$$G/\operatorname{Rad}_{\mathrm{MF}}(G) \cong Q/\operatorname{Rad}_{\mathrm{MF}}(Q).$$

These correspondences respect composition of surjective homomorphisms with kernels invisible to MF.

Let B be a countable group with $\operatorname{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \ne d \in B$ normally generate B. Put $A = \langle d\rangle$. For a countable group Q, define

$$W_Q = B *_A (Q \times A), \tag{11}$$

where $A \to Q \times A$ is the inclusion into the second factor. The trivial map $B \to Q$ and the projection $Q \times A \to Q$ agree on A. They therefore define an epimorphism

$$\pi_Q\colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (11) are injective; in particular, $d \ne 1$ in W_Q.

Proposition 7.2 (universal factorization)**.** *Let T be a group such that every homomorphism $B \to T$ is trivial. Then precomposition with π_Q is a bijection*

$$\operatorname{Hom}(Q, T) \longrightarrow \operatorname{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

Proof. Let $f\colon W_Q \to T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q, a) \in Q \times A$, we then have

$$f(q, a) = f(q, 1)f(1, a) = f(q, 1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\!\langle d \rangle\!\rangle_{W_Q}$ kills B, since $\langle\!\langle d \rangle\!\rangle_B = B$, and kills the A-factor of $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}$. □

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\operatorname{Rad}_{\mathrm{MF}}(B) = B$. Proposition 7.2 therefore gives the asserted bijection for every MF group M. In addition, the definition of $\operatorname{Rad}_{\mathrm{MF}}(B) = B$ says directly that every homomorphism from B to $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is trivial. Applying Proposition 7.2 with $T = \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ shows that every corona homomorphism from W_Q factors uniquely through π_Q. Intersecting their kernels gives

$$\operatorname{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\operatorname{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\operatorname{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

□

If $N \trianglelefteq W_Q$, the same factorization gives

$$\operatorname{cl}_{\mathrm{MF}}^{W_Q}(N) = \pi_Q^{-1}\left(\operatorname{cl}_{\mathrm{MF}}^{Q}(\pi_Q(N))\right). \tag{12}$$

Indeed, a homomorphism from W_Q to an MF group kills N exactly when its unique factor through Q kills $\pi_Q(N)$. Proposition 1.1 and (12) give

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \le N \text{ and } Q/\pi_Q(N) \text{ is MF}.$$

Here, if $\ker \pi_Q \le N$, then $N = \pi_Q^{-1}(\pi_Q(N))$.

Taking $Q = \mathbb{Z}$ makes the quotient criterion especially concrete: for every normal subgroup $N \trianglelefteq W_{\mathbb{Z}}$,

$$W_{\mathbb{Z}}/N \text{ is MF} \iff d \in N. \tag{13}$$

Indeed, every quotient of $\mathbb{Z}$ is MF, while $\ker \pi_{\mathbb{Z}} = \langle\!\langle d \rangle\!\rangle_{W_{\mathbb{Z}}}$.

8. Operator norm models on finite sets

The recognition of MF groups from finite presentations, taken up in the following sections, rests on the observation that the MF property, although defined through sequences of models on all of G, is determined by finite data with a fixed separation constant. This section isolates that reformulation.

Lemma 8.1 (locality with fixed separation)**.** *A countable group G is MF if and only if for every finite set $F \subseteq G$ containing 1 and every $\varepsilon > 0$ there are $d \geq 1$ and a map $V : F \to \mathcal{U}(d)$ with $V(1) = 1$ such that*

$$\|V(gh) - V(g)V(h)\| \leq \varepsilon \quad \textit{whenever } g, h, gh \in F,$$
$$\|V(g) - 1\| \geq 1 \quad \textit{for } g \in F \setminus \{1\}.$$

Proof. Suppose first that every finite $F \ni 1$ and every $\varepsilon > 0$ admit such a map. Enumerate $G = \{g_0 = 1, g_1, g_2, \dots\}$, let $F_n = \{g_i g_j : i, j \leq n\} \cup \{g_i : i \leq n\}$, let $V_n : F_n \to \mathcal{U}(d_n)$ be a map for $(F_n, 1/n)$, and extend it to G by $V_n(g) = 1$ for $g \notin F_n$. For $g, h \in G$ the elements g, h, gh lie in F_n for all large n, so $\|V_n(gh) - V_n(g)V_n(h)\| \leq 1/n \to 0$; and for $g \neq 1$ we have $\|V_n(g) - 1\| \geq 1$ for all large n. Hence G is MF.

Conversely, let $V_n : G \to \mathcal{U}(d_n)$ be MF models: $V_n(1) = 1$, the multiplication defects tend to zero, and $c_g = \limsup_n \|V_n(g) - 1\| > 0$ for every $g \neq 1$. The separation constants c_g may be arbitrarily small, and we first amplify them. For a unitary u and $k \geq 1$ the tensor power $u^{\otimes k}$ is a unitary of dimension d^k, $(uv)^{\otimes k} = u^{\otimes k} v^{\otimes k}$, and telescoping gives $\left\| u^{\otimes k} - v^{\otimes k} \right\| \leq k \|u - v\|$. If $\|u - 1\| = c > 0$, write $c = 2\sin(\theta/2)$ with $0 < \theta \leq \pi$; then $e^{i\theta}$ or $e^{-i\theta}$ is an eigenvalue of u, so $e^{ik\theta}$ or $e^{-ik\theta}$ is an eigenvalue of $u^{\otimes k}$, and $\left\| u^{\otimes k} - 1 \right\| \geq |e^{ik\theta} - 1| = 2|\sin(k\theta/2)|$. If $\theta \geq \pi/2$ take $k = 1$; otherwise the first k with $k\theta \geq \pi/2$ satisfies $k\theta < \pi/2 + \theta < \pi$. In both cases $k\theta \in [\pi/2, 3\pi/2]$ modulo 2π, so $\left\| u^{\otimes k} - 1 \right\| \geq \sqrt{2}$. Write $k(c)$ for this exponent; it depends only on c.

Now fix a finite $F \ni 1$ and $\varepsilon > 0$, let $K = \max_{g \in F \setminus \{1\}} k(c_g)$, and put

$$W_n(g) = \bigoplus_{k=1}^{K} V_n(g)^{\otimes k} \in \mathcal{U}(D_n), \qquad D_n = \sum_{k=1}^{K} d_n^k.$$

Then $W_n(1) = 1$ and $\|W_n(gh) - W_n(g)W_n(h)\| \leq K \|V_n(gh) - V_n(g)V_n(h)\|$, so the defects of W_n tend to zero. For $g \in F \setminus \{1\}$ choose a subsequence along which $\|V_n(g) - 1\| \to c_g$ and eigenvalues λ_n of $V_n(g)$ with $|\lambda_n - 1| \to c_g$; passing to a further subsequence, $\lambda_n \to \lambda$ with $|\lambda - 1| = c_g$, and then $\lambda_n^k \to \lambda^k$ with $|\lambda^k - 1| \geq \sqrt{2}$ for $k = k(c_g)$. Since λ_n^k is an eigenvalue of $V_n(g)^{\otimes k}$, it follows that $\limsup_n \|W_n(g) - 1\| \geq \sqrt{2} > 1$. Choose n_0 such that for $n \geq n_0$ every defect of W_n on F is at most ε, and for each $g \in F \setminus \{1\}$ choose $m_g \geq n_0$ with $\|W_{m_g}(g) - 1\| \geq 1$. Let V be the restriction to F of $\bigoplus_{g \in F \setminus \{1\}} W_{m_g}$ (and of W_{n_0} if $F = \{1\}$). Then $V(1) = 1$, the defect of V

on F is the maximum of the defects of its blocks, hence at most ε, and for $g \in F \setminus \{1\}$ the block m_g gives $\|V(g) - 1\| \geq 1$. $\square$

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set $A \subseteq \mathbb{N}$ is Σ^0_1 if $A = \{x : \exists y\, R(x, y)\}$ for a decidable relation R, it is Π^0_1 if its complement is Σ^0_1, and Σ^0_2 and Π^0_2 are the classes $\{x : \exists y\, \forall z\, R(x, y, z)\}$ and $\{x : \forall y\, \exists z\, R(x, y, z)\}$ with R decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_P for the group presented by the code P. Let $\mathrm{MF}_{\mathrm{fp}}$ be the set of codes P such that G_P is MF, and $\mathrm{NONMF}_{\mathrm{fp}}$ its complement.

Proposition 8.2 (the certificate normal form)**.** *There is a decidable relation $C(P, n, c)$ on triples of natural numbers such that a code P lies in $\mathrm{MF}_{\mathrm{fp}}$ if and only if for every n there is c with $C(P, n, c)$. Consequently $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ and $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$.*

Proof. Let $P = \langle x_1, \dots, x_k \mid r_1, \dots, r_m\rangle$. A *certificate at scale n* is a triple $c = (d, \ell, \pi)$ consisting of a dimension $d \geq 1$, a labelling ℓ of every reduced word of length at most n in $x_1^{\pm 1}, \dots, x_k^{\pm 1}$ by one of the symbols T and S, and, for every word w labelled T, an expression of w as a product of conjugates of the relators and their inverses that freely reduces to w. Such an expression is checkable, and its existence forces $w = 1$ in G_P. Let $\Phi(P, n, c)$ be the statement that there are $U_1, \dots, U_k \in \mathcal{U}(d)$ with $\|r_i(U) - 1\| \leq 2^{-n}$ for $i \leq m$ and $\|w(U) - 1\| \geq 1/4$ for every word w labelled S, where $w(U)$ is the evaluation of w at $U_1, \dots, U_k$ with $x_j^{-1} \mapsto U_j^*$. Writing $U_j = A_j + iB_j$ with real matrices, unitarity is a system of polynomial equations, and a bound $\|X\| \leq t$ or $\|X\| \geq t$ on a matrix whose entries are polynomials in the variables is a semialgebraic condition, since $\|X\| \leq t$ holds precisely when $t^2 - X^*X$ is positive semidefinite. Thus $\Phi(P, n, c)$ is a sentence of the first-order theory of the ordered field of real numbers, and its truth is decidable by Tarski's theorem [Ta]. Let $C(P, n, c)$ hold when c is a well-formed certificate at scale n whose T-expressions check and $\Phi(P, n, c)$ holds. Then C is decidable.

Suppose that G_P is MF. Fix n, let $L = \max(n, |r_1|, \dots, |r_m|)$, let F be the set of images in G_P of all words of length at most L, together with their inverses, and let $\varepsilon = 2^{-n}/(4L)$. Lemma 8.1 gives $V\colon F \to \mathcal{U}(d)$ with $V(1) = 1$, defect at most ε on F, and $\|V(g) - 1\| \geq 1$ for $g \in F \setminus \{1\}$. Put $U_j = V(\bar{x}_j)$, where $\bar{w}$ denotes the image of a word w in G_P. For $g \in F$ we have $g^{-1} \in F$ and $\|V(g)V(g^{-1}) - 1\| \leq \varepsilon$, so $\|V(g^{-1}) - V(g)^*\| \leq \varepsilon$. For a word $w = y_1 \cdots y_\ell$ of length $\ell \leq L$, all of whose prefixes have images in F, induction on ℓ gives $\|w(U) - V(\bar{w})\| \leq 2\ell\varepsilon$. In particular $\|r_i(U) - 1\| \leq 2L\varepsilon \leq 2^{-n}$. Label a word w of length at most n by T if $\bar{w} = 1$, with a normal-closure expression as witness, and by S otherwise; for the latter, $\|w(U) - 1\| \geq \|V(\bar{w}) - 1\| - 2n\varepsilon \geq 1 - \frac{1}{2} \geq \frac{1}{4}$. So $C(P, n, c)$ holds for this certificate.

Conversely, suppose that for every n some certificate $c_n = (d_n, \ell_n, \pi_n)$ satisfies $C(P, n, c_n)$, and choose $U^{(n)} \in \mathcal{U}(d_n)^k$ witnessing $\Phi(P, n, c_n)$. For each $g \in G_P$ fix a word w_g representing g, with w_1 the empty word, and put $V_n(g) = w_g(U^{(n)})$. Then $V_n(1) = 1$. For $g, h \in G_P$ the word $t = w_g w_h w_{gh}^{-1}$ is trivial in G_P, so it is freely equal to a product of A conjugates of relators and their inverses, for some A depending on g and h but not on n. Since evaluation is a homomorphism on the free group, since the operator norm is unitarily invariant, and since $\|xy - 1\| \leq \|x - 1\| + \|y - 1\|$ for unitaries x, y,

$$\|V_n(g)V_n(h) - V_n(gh)\| = \left\|t(U^{(n)}) - 1\right\| \leq A\,2^{-n} \longrightarrow 0.$$

For $g \neq 1$ the word w_g is nontrivial in G_P, so for $n \geq |w_g|$ the certificate c_n cannot label it T, hence labels it S, and $\|V_n(g) - 1\| \geq 1/4$. Thus $\limsup_n \|V_n(g) - 1\| \geq 1/4 > 0$, and G_P is MF.

The displayed equivalence exhibits $\mathrm{MF}_{\mathrm{fp}}$ as $\{P : \forall n\, \exists c\, C(P, n, c)\}$, a Π^0_2 set, and its complement as a Σ^0_2 set. □

9. HNN extensions with a corona conjugator

The lower bound in Theorem D is a computable family of finitely presented groups whose positive branch must be MF. Those groups are built by two HNN extensions, and the tool that keeps them MF is the following permanence theorem. It asks for more than the MF property of the base: a realization carrying a trace that is regular on the group.

A countable group G is *regularly realized* by a triple (A, ρ, τ) if A is a separable unital MF C^*-algebra, $\rho\colon G \to \mathcal{U}(A)$ is a homomorphism, and τ is a tracial state on A with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, a regularly realized group embeds in $\mathcal{U}(A)$, so it is MF. Padding blocks with zeros embeds every norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ isometrically into $\prod_n M_n(\mathbb{C})/\bigoplus_n M_n(\mathbb{C})$, so the MF algebras of the introduction are exactly the separable C^*-algebras that embed in the latter corona, as in [BK, Sh].

Lemma 9.1 (residually finite groups are regularly realized)**.** *Every countable residually finite group is regularly realized.*

Proof. Let G be countable and residually finite, and choose finite-index normal subgroups $N_1 \geq N_2 \geq \cdots$ with $\bigcap_k N_k = \{1\}$; this is possible because G is countable. Let $d_k = [G : N_k]$ and let $\lambda_k\colon G \to \mathcal{U}(d_k)$ be the left regular representation of G/N_k. In the corona $\mathcal{Q}_{\mathbf{d}}$ put $\rho(g) = [(\lambda_k(g))_k]$; then ρ is a homomorphism into the unitary group. Fix a free ultrafilter ω on $\mathbb{N}$ and set $\tau([(x_k)_k]) = \lim_\omega \mathrm{tr}_{d_k}(x_k)$, where tr_{d_k} is the normalized trace on $M_{d_k}(\mathbb{C})$; this is well defined because $|\mathrm{tr}_{d_k}(x_k)| \leq \|x_k\|$, and it is a tracial state on $\mathcal{Q}_{\mathbf{d}}$. If $g \neq 1$, then $g \notin N_k$ for all large k, so $\mathrm{tr}_{d_k}(\lambda_k(g)) = 0$ for all large k and $\tau(\rho(g)) = 0$. Let A be the C^*-algebra generated by $\rho(G)$ and the unit of $\mathcal{Q}_{\mathbf{d}}$; it is separable, unital, and MF, and $(A, \rho, \tau|_A)$ is a regular realization of G. □

Let G be a countable group, let $S \leq G$ be a subgroup, and let $\theta\colon S \to G$ be an injective homomorphism. The HNN extension

$$R = \langle G, t \mid tst^{-1} = \theta(s)\ (s \in S)\rangle \tag{14}$$

contains G by Britton's lemma [Bri].

Theorem 9.2 (HNN permanence with a corona conjugator)**.** *In the situation of* (14)*, suppose that G is regularly realized by (A, ρ, τ), and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$-homomorphism $\iota\colon A \to \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W\,\iota\rho(s)\,W^* = \iota\rho(\theta(s)) \qquad (s \in S).$$

Then R is regularly realized. In particular, R is MF.

Proof. Write $D = C^*(\iota\rho(G)) \subseteq \mathcal{Q}$, a separable C^*-algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota\rho(S))$ and $B_1 = C^*(\iota\rho(\theta(S)))$, both containing p. Conjugation by W restricts to a $*$-isomorphism $\Theta\colon B_0 \to B_1$ carrying $\iota\rho(s)$ to $\iota\rho(\theta(s))$. Let U be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where u is the canonical unitary generator of $C(\mathbb{T})$. We write d and u for the images of $d \in D$ and u in U. By construction, U has the following universal property: whenever $\pi\colon D \to E$ is a unital $*$-homomorphism into a unital C^*-algebra and $v \in \mathcal{U}(E)$ satisfies $v\pi(b)v^* = \pi(\Theta(b))$ for $b \in B_0$, there is a unique $*$-homomorphism $U \to E$ extending π and sending u to v. The proof has three steps: U embeds in an amalgamated free product; that amalgamated free product is MF; and R embeds in $\mathcal{U}(U)$ with a regular trace.

Step 1: U embeds in an amalgamated free product. This step is a self-contained form of a corner argument of Ueda [Ue]. Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \qquad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which A_1 and A_2 embed [Sh, Section 2.1]. Write e_{ij} and f_{ij} for the matrix units of A_1 and A_2. Since $\iota_A(p, 0) = \iota_B(p, 0)$ and $\iota_A(0, p) = \iota_B(0, p)$, we have $e_{11} = f_{11} =: e$ and $e_{22} = f_{22}$ in P. Define $\pi_D\colon D \to ePe$ by $\pi_D(d) = \operatorname{diag}(d, 0)$, a unital $*$-homomorphism into the corner, and put $w = e_{12}f_{21} \in ePe$. Then $ww^* = e_{12}f_{22}e_{21} = e_{12}e_{22}e_{21} = e$ and $w^*w = f_{12}e_{22}f_{21} = f_{12}f_{22}f_{21} = e$, so w is a unitary of ePe. For $b \in B_0$ the element $\operatorname{diag}(b, 0)$ is $\iota_A(b, 0) = \iota_B(b, 0)$, so

$$\begin{aligned} w\pi_D(b)w^* &= e_{12}\big(f_{21}\operatorname{diag}(b, 0)f_{12}\big)e_{21} = e_{12}\,\iota_B(0, \Theta(b))\,e_{21} \\ &= e_{12}\,\iota_A(0, \Theta(b))\,e_{21} = \pi_D(\Theta(b)), \end{aligned}$$

where the middle equality uses $\iota_B(0, \Theta(b)) = \operatorname{diag}(0, b)$ in A_2. The universal property of U gives a $*$-homomorphism $\Phi\colon U \to ePe \subseteq P$ with $\Phi|_D = \pi_D$ and $\Phi(u) = w$. To see that Φ is injective, map back. Let $\Psi_1\colon A_1 \to M_2(U)$ be the entrywise inclusion $D \subseteq U$, and let $\Psi_2\colon A_2 \to M_2(U)$ be the entrywise inclusion $B_0 \subseteq U$ followed by conjugation by $\operatorname{diag}(1, u)$. On C we have

$\Psi_1(\iota_A(c_0,c_1)) = \operatorname{diag}(c_0,c_1)$ and $\Psi_2(\iota_B(c_0,c_1)) = \operatorname{diag}(c_0, u\Theta^{-1}(c_1)u^*) = \operatorname{diag}(c_0,c_1)$, so the pair induces a $*$-homomorphism $\Psi\colon P \to M_2(U)$. On generators, $\Psi\Phi(d) = \operatorname{diag}(d,0)$ and $\Psi\Phi(u) = \Psi_1(e_{12})\Psi_2(f_{21}) = \operatorname{diag}(u,0)$. Hence $\Psi\Phi$ is the injective $*$-homomorphism $x \mapsto \operatorname{diag}(x,0)$ of U into $M_2(U)$, and Φ is injective.

Step 2: P is MF. By Shulman's criterion [Sh, Theorem 16], the full amalgamated free product $A_1 *_C A_2$ of separable C^*-algebras is MF as soon as there are injective $*$-homomorphisms φ_A and φ_B of A_1 and A_2 into one norm matrix corona with $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$. Identify $M_2(\mathcal{Q})$ with the norm matrix corona of the doubled dimension sequence, and let $\varphi_A\colon M_2(D) \to M_2(\mathcal{Q})$ be the entrywise inclusion and $\varphi_B\colon M_2(B_0) \to M_2(\mathcal{Q})$ the entrywise inclusion followed by conjugation by $\operatorname{diag}(1,W)$. Both are injective, and on C,

$$\varphi_B(\iota_B(c_0,c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0,c_1)),$$

because Θ is conjugation by W. Hence P is MF, and so are its C^*-subalgebras ePe and $\Phi(U) \cong U$.

Step 3: R embeds in $\mathcal{U}(U)$ with a regular trace. The functional $T = \tau \circ \iota^{-1}$ is a tracial state on $\iota(A) \supseteq D$ with $T(\iota\rho(g)) = 0$ for $g \neq 1$. In the GNS representation of D for T, the vectors $\xi_g = \iota\rho(g)\xi_T$, $g \in G$, are orthonormal, because $\langle \xi_g, \xi_h\rangle = T(\iota\rho(h^{-1}g))$. Their closed span is invariant under D, and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of G. Thus there is a $*$-homomorphism $\pi\colon D \to C^*_{\mathrm{r}}(G)$ with $\pi(\iota\rho(g)) = \lambda_G(g)$. Since $G \leq R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of G, so $\lambda_R|_G$ is a multiple of λ_G and the assignment $\lambda_G(g) \mapsto \lambda_R(g)$ extends to a $*$-homomorphism $C^*_{\mathrm{r}}(G) \to C^*_{\mathrm{r}}(R)$. Composing, we obtain $\sigma_0\colon D \to C^*_{\mathrm{r}}(R)$ with $\sigma_0(\iota\rho(g)) = \lambda_R(g)$. The pair $(\sigma_0, \lambda_R(t))$ satisfies the covariance relation $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in R, and two $*$-homomorphisms that agree on generators agree on B_0. The universal property of U gives $\sigma\colon U \to C^*_{\mathrm{r}}(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j\colon R \to \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (14) hold in U, since $u\,\iota\rho(s)\,u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so j is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so j is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF C^*-algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where τ_R is the canonical trace $x \mapsto \langle x\delta_1, \delta_1\rangle$ of $C^*_{\mathrm{r}}(R)$. Then τ' is a tracial state on A' with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. Hence (A', j, τ') regularly realizes R. □

Nothing in the proof asserts that the concrete pair $(\iota\rho, W)$ is a faithful representation of R in $\mathcal{Q}$; in the applications below it is not. Faithfulness is bought at the universal algebra U by the regular trace of the base together with the left regular representation of R itself, while the operator norm approximation is bought by Shulman's criterion. No reduced normal form for the HNN extension and no freeness estimate appears.

Corollary 9.3 (central HNN extensions)**.** *If G is regularly realized and $S \le G$ is any subgroup, then $\langle G, t \mid [t,s] = 1\ (s \in S)\rangle$ is regularly realized.*

Proof. Apply Theorem 9.2 with θ the inclusion of S, with ι any injective $*$-homomorphism of A into a norm matrix corona, and with $W = 1$. □

10. Compiling the halting hierarchy into finite presentations

Write W_e for the domain of the e-th partial computable function and

$$\mathrm{INF} = \{e : W_e \text{ is infinite}\}, \qquad \mathrm{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is Π^0_2-complete and FIN is Σ^0_2-complete [So, Chapter IV]. This section constructs a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the presented group is MF if and only if $e \in \mathrm{INF}$. The construction has three moves. A *switch* turns e into a recursively presented group that is trivial for $e \in \mathrm{INF}$ and contains H for $e \in \mathrm{FIN}$. A *rope*, in the manner of Higman's embedding theorem, wraps the switch into a finitely presented group by two HNN extensions. A *synchronization* of finite quotients then supplies the corona conjugator that Theorem 9.2 needs on the positive branch. The first two moves are group theory and occupy this section up to Lemma 10.7; the third is carried out in the next section.

The seed.

Lemma 10.1 (the seed is recursively presented)**.** *The group $H = \mathrm{EL}_{12}(R)$ is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

Proof. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i, j, k, and since every element of R is a sum of monomials in the generators s_0, s_1, t_0, t_1, induction on the length of a monomial shows that every $e_{ij}(a)$ lies in the subgroup generated by Y; there is always an index $k \notin \{i, j\}$ because $12 \ge 3$. Thus $H = \langle Y \rangle$. The monomials in s_0, s_1, t_0, t_1 containing no factor $s_1 t_1$ form an $\mathbb{F}_2$-basis of R [Lev], and the product of two basis monomials is computed from the defining relations by finitely many rewritings, so equality in R is decidable. A word in $Y^{\pm 1}$ is therefore evaluated to an explicit 12×12 matrix over R, and it represents 1 in H exactly when that matrix is the identity. The set of words representing 1 is decidable, hence recursively enumerable, and $\langle Y \mid \text{that set}\rangle$ is a recursive presentation of H. □

The switch.

Lemma 10.2 (the switch)**.** *There is an algorithm that computes from e a recursive presentation, on a computable set of generators, of a group C_e such that C_e is trivial if $e \in \mathrm{INF}$ and H embeds in C_e if $e \in \mathrm{FIN}$.*

Proof. This is the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] with the trivial group as positive witness and H as negative witness. Let $\langle Y \mid T\rangle$ be the recursive presentation of H from Lemma 10.1. Take generators $y_{i,\ell}$ for $i \geq 1$ and $\ell \in Y$, one copy Y_i of Y for each i, and as relators all words of T rewritten in each copy Y_i, together with every generator $y_{i,\ell}$ for which $i \leq |W_e|$. The last family is recursively enumerable uniformly in e: enumerate W_e, and whenever its i-th element appears, enumerate all of Y_i. The group presented is the free product of the copies $H_i \cong H$, $i \geq 1$, with the first $|W_e|$ copies killed. If W_e is infinite every copy is killed and C_e is trivial; if W_e is finite, C_e is the free product of the remaining copies and contains H. $\square$

The bridge. Let $F = F(x, y, t)$ be the free group on x, y, t, and let $y_i = x^i y x^{-i}$ for $i \in \mathbb{Z}$. For a countable group C with a generating sequence $(c_i)_{i\geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$B(C) = \langle C * F(x, y),\ t \ \mid\ t y_i t^{-1} = c_i y_i\ (i \in \mathbb{Z})\rangle. \tag{15}$$

Let $H_0 = B(1) = \langle x, y, t \mid [t, y_i] = 1\ (i \in \mathbb{Z})\rangle$, let $q_+ \colon F \to H_0$ be the quotient map, and let $N_+ = \ker q_+$.

Lemma 10.3 (the bridge)**.**

(1) $B(C)$ is an HNN extension of $C * F(x, y)$, so C embeds in $B(C)$; it is generated by x, y, t; and a recursive presentation of $B(C)$ on x, y, t is computable from a recursive presentation of C. We write $B(C) = F/N_C$ and $q_C \colon F \to B(C)$ for the quotient map.

(2) Killing C induces a surjection $B(C) \to H_0$; hence $N_C \leq N_+$, with equality when C is trivial.

(3) The assignment $x \mapsto (x_1, x_2)$, $y \mapsto (y, 1)$, $t \mapsto (1, t)$ defines an injective homomorphism $j \colon H_0 \to P = F(x_1, y) \times F(x_2, t)$. In particular, H_0 is residually finite.

Proof. (1) The elements y_i, $i \in \mathbb{Z}$, freely generate the normal closure of y in $F(x, y)$ [LS, Chapter I]. The retraction $C * F(x, y) \to F(x, y)$ killing C sends $c_i y_i$ to y_i, so a nontrivial reduced word in the $c_i y_i$ maps to a nontrivial word in the y_i; hence the $c_i y_i$ freely generate a free subgroup, and $y_i \mapsto c_i y_i$ is an isomorphism between two free subgroups of the base. Thus (15) is an HNN extension, and the base, hence C, embeds by Britton's lemma. The relation $c_i = t y_i t^{-1} y_i^{-1}$ shows that x, y, t generate $B(C)$, and substituting this expression for c_i into the relators of C and into (15) produces a recursive presentation on x, y, t, uniformly in the presentation of C.

(2) The relators of (15) map to the relators $[t, y_i]$ of H_0 when every c_i is sent to 1, and the map is the identity on x, y, t.

(3) The relators $[t, y_i]$ are sent to $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$, so j is a homomorphism. Let N_0 be the normal closure of $\{y, t\}$ in H_0. Then $H_0/N_0 = \langle x\rangle$ is infinite cyclic and the extension splits, so $H_0 = N_0 \rtimes \langle x\rangle$. Rewriting the presentation of H_0 over the transversal $\{x^n\}$ by the Reidemeister–Schreier method [LS,

Chapter II] presents N_0 on the generators $y_n = x^n y x^{-n}$ and $t_m = x^m t x^{-m}$, $n, m \in \mathbb{Z}$, with relators the conjugates $x^m[t, y_n]x^{-m} = [t_m, y_{n+m}]$, that is, all commutators $[t_m, y_n]$. Hence $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$. Now $j(y_n) = (x_1^n y x_1^{-n}, 1)$ and $j(t_m) = (1, x_2^m t x_2^{-m})$, and the elements $x_1^n y x_1^{-n}$ freely generate the normal closure of y in $F(x_1, y)$, and likewise for t; so j restricted to N_0 is an isomorphism onto the product of these two normal closures. Finally $j(x) = (x_1, x_2)$ has infinite order and $j(x)^k \notin j(N_0)$ for $k \neq 0$, because the exponent sum of x_1 in $j(x^k n)$ is k for $n \in N_0$. Hence j is injective on $N_0 \rtimes \langle x \rangle$. Free groups are residually finite and residual finiteness passes to direct products and subgroups, so H_0 is residually finite. $\square$

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with C_e the group of Lemma 10.2, and write $q_e \colon F \to Q_e$. By Lemmas 10.2 and 10.3, a recursive presentation of Q_e on x, y, t is computable from e; $N_e \leq N_+$; if $e \in \mathrm{INF}$ then $Q_e = H_0$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \mathrm{FIN}$ then $H \leq C_e \leq Q_e$.

The cutting subgroup.

Lemma 10.4 (Higman embedding and the Mikhailova subgroup)**.** *There is an algorithm that computes from e a finite presentation $\langle X_e \mid R_e \rangle$ of a group $\widehat{H}_e$, words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup $M_e \leq F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \to \widehat{H}_e$, and $M_e = \{(u, v) : u = v \text{ in } \widehat{H}_e\}$. Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \qquad i_e(f) = (f, w_f, 1), \qquad L_e^0 = F \times M_e,$$

where w_f is the image of f under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e \colon F \to K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.

Proof. Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of Q_e. Following Mikhailova [Mih], let M_e be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs (z, z), $z \in X_e$, and $(r, 1)$, $r \in R_e$. If $(u, v) \in M_e$ then $u = v$ in $\widehat{H}_e$, since this holds for the generators. Conversely, if $u = v$ in $\widehat{H}_e$, then uv^{-1} lies in the normal closure of R_e, so it is a product of conjugates $gr^{\pm 1}g^{-1}$ with $r \in R_e$, and $(gr^{\pm 1}g^{-1}, 1) = (g, g)(r^{\pm 1}, 1)(g, g)^{-1} \in M_e$; hence $(u, v) = (uv^{-1}, 1)(v, v) \in M_e$. The first coordinate makes i_e injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in $\widehat{H}_e$, if and only if $q_e(f) = 1$ because the embedding is injective, if and only if $f \in N_e$. $\square$

The ropes. Let $P = F(x_1, y) \times F(x_2, t)$ and $j \colon H_0 \to P$ be as in Lemma 10.3. Define the graph witness

$$K^{\mathrm{g}} = F \times P, \qquad L^{\mathrm{g}} = \{(f,\, jq_+(f)) : f \in F\},$$

and the product witness

$$K_e = K_e^0 \times K^{\mathrm{g}}, \qquad L_e = L_e^0 \times L^{\mathrm{g}}, \qquad i\colon F \to K_e, \quad i(f) = \big(i_e(f), (f,1)\big).$$

Let

$$\Gamma_e = \big\langle K_e,\ v \ \big|\ [v,\ell] = 1\ (\ell \in L_e)\big\rangle, \qquad S_e = \big\langle i(F),\ v\, i(F)\, v^{-1}\big\rangle \le \Gamma_e. \tag{16}$$

Lemma 10.5 (the central rope)**.**

(1) K_e is a direct product of free groups of finite rank, hence finitely presented and residually finite; L_e is finitely generated; i is injective; and $i(F) \cap L_e = i(N_e)$.

(2) Γ_e is finitely presented, and a finite presentation is computable from e. The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with n in the second copy, onto S_e, sending the first copy to $i(F)$ and the second to $v\, i(F)\, v^{-1}$, is an isomorphism.

(3) There is a homomorphism $\tau_e\colon S_e \to Q_e$ with $\tau_e(i(f)) = q_e(f)$ and $\tau_e(v\, i(f)\, v^{-1}) = 1$ for $f \in F$.

Proof. (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of five free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup L_e is generated by the basis of the first factor, the generators of M_e from Lemma 10.4, and the elements $(a, jq_+(a))$ for $a \in \{x, y, t\}$, so it is finitely generated. Injectivity of i is clear from the first coordinate. An element $i(f)$ lies in L_e if and only if $i_e(f) \in L_e^0$ and $(f,1) \in L^{\mathrm{g}}$; the first holds if and only if $f \in N_e$ by Lemma 10.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \le N_+$, the intersection is $i(N_e)$.

(2) The presentation (16) is finite once $[v,\ell] = 1$ is imposed only for the finitely many generators ℓ of L_e, and it is computable from e because K_e, L_e and i are. It is an HNN extension of K_e with both edge groups equal to L_e and the identity as edge isomorphism. The map from $F *_{N_e} F$ to S_e is well defined because $i(n) \in L_e$ commutes with v for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two copies of F lying outside N_e; its image is a word $i(f_0)\, v\, i(f_1)\, v^{-1}\, i(f_2) \cdots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, hence $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in Γ_e by Britton's lemma.

(3) Define τ_e on $F *_{N_e} F$ by q_e on the first copy and by the trivial map on the second; both kill N_e, so the definition is consistent on the amalgamated subgroup, and (2) transports it to S_e. □

The second rope is the HNN extension

$$R_e = \big\langle \Gamma_e \times Q_e,\ u \ \big|\ u(s,1)u^{-1} = (s, \tau_e(s))\ (s \in S_e)\big\rangle, \tag{17}$$

whose edge maps $s \mapsto (s,1)$ and $s \mapsto (s, \tau_e(s))$ are injective homomorphisms of S_e into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in R_e by Britton's lemma. The presentation (17) lists the recursively enumerable relators of Q_e. Its finite replacement is

(18)
$$\widehat{R}_e = \Big\langle \Gamma_e \times F,\ u \ \Big|\ u(i(a),1)u^{-1} = (i(a),a), \quad u(v\,i(a)\,v^{-1},1)u^{-1} = (v\,i(a)\,v^{-1},1) \quad (a \in \{x,y,t\}) \Big\rangle,$$

where $\Gamma_e \times F$ carries the finite presentation of Lemma 10.5(2), the basis x, y, t of F, and the commutation relators between the two factors.

Lemma 10.6 (the finite presentation)**.** *The groups $\widehat{R}_e$ and R_e are isomorphic by the map that is the identity on Γ_e and on u and sends F onto Q_e by q_e. A code of the finite presentation (18) is computable from e.*

Proof. Both maps $f \mapsto u(i(f),1)u^{-1}$ and $f \mapsto (i(f),f)$ are homomorphisms $F \to \widehat{R}_e$, and they agree on the basis, so $u(i(f),1)u^{-1} = (i(f),f)$ for all $f \in F$; likewise $u(v\,i(f)\,v^{-1},1)u^{-1} = (v\,i(f)\,v^{-1},1)$ for all f. If $n \in N_e$, then $i(n) \in L_e$ by Lemma 10.5(1), so $v\,i(n)\,v^{-1} = i(n)$ in Γ_e, and comparing the two relations gives $(i(n),n) = (i(n),1)$, that is, $(1,n) = 1$ in $\widehat{R}_e$. Hence the factor F of $\widehat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (18) become $u(s,1)u^{-1} = (s,\tau_e(s))$ for s in the generating set $\{i(a),\ v\,i(a)\,v^{-1} : a \in \{x,y,t\}\}$ of S_e, hence for all $s \in S_e$, since both sides are homomorphic in s. This defines a homomorphism $R_e \to \widehat{R}_e$; conversely, the relations of (18) hold in R_e after $F \to Q_e$, because $\tau_e(i(a)) = q_e(a)$ and $\tau_e(v\,i(a)\,v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 10.4 and 10.5. ☐

Lemma 10.7 (the negative branch)**.** *If $e \in \mathrm{FIN}$, then H embeds in $\widehat{R}_e$, and $\widehat{R}_e$ is not MF.*

Proof. For $e \in \mathrm{FIN}$ we have $H \le C_e \le Q_e \le \Gamma_e \times Q_e \le R_e \cong \widehat{R}_e$ by Lemmas 10.2, 10.3(1), 10.6 and Britton's lemma for (17). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and H is not MF by Theorem B. ☐

11. The positive branch

We now show that $\widehat{R}_e$ is MF for $e \in \mathrm{INF}$ and prove Theorem D. The realization of the rope is assembled inside a norm reduced product of MF algebras, so we first record that such products stay MF. For C^*-algebras B_n, $n \ge 1$, write $\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \to 0$; its norm is $\limsup_n \|x_n\|$.

Lemma 11.1 (reduced products of MF algebras)**.** *If every B_n is MF, then every separable C^*-subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever B is.*

Proof. The second assertion holds because M_k of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by k. For the first, let A be a separable C^*-subalgebra, and choose a sequence $(a_i)_{i\geq 1}$ dense in A, with lifts $(a_i^{(n)})_n$. Choose for each n an injective $*$-homomorphism η_n of B_n into a norm matrix corona, and lifts $(a_i^{(n,k)})_k$ of $\eta_n(a_i^{(n)})$ with $\sup_k \left\|a_i^{(n,k)}\right\| \leq \left\|a_i^{(n)}\right\| + 1$. Enumerate the noncommutative $*$-polynomials with coefficients in $\mathbb{Q}(i)$ as $p_1, p_2, \ldots$, and let $\mathcal{P}_n$ be the set of those among $p_1, \ldots, p_n$ that involve only $a_1, \ldots, a_n$. For $p \in \mathcal{P}_n$ the number $\left\|p(a^{(n)})\right\|$ equals $\limsup_k \left\|p(a^{(n,k)})\right\|$, since η_n is isometric. Hence there is K_n such that $\left\|p(a^{(n,k)})\right\| \leq \left\|p(a^{(n)})\right\| + 1/n$ for all $k \geq K_n$ and $p \in \mathcal{P}_n$, and for each $p \in \mathcal{P}_n$ there is $k_p \geq K_n$ with $\left\|p(a^{(n,k_p)})\right\| \geq \left\|p(a^{(n)})\right\| - 1/n$. Put

$$b_i^{(n)} = \bigoplus_{p\in\mathcal{P}_n} a_i^{(n,k_p)},$$

a block diagonal matrix. For $p \in \mathcal{P}_n$ the norm of $p(b^{(n)})$ is the maximum of $\left\|p(a^{(n,k_{p'})})\right\|$ over $p' \in \mathcal{P}_n$, which lies between $\left\|p(a^{(n)})\right\| - 1/n$ and $\left\|p(a^{(n)})\right\| + 1/n$. Every polynomial belongs to $\mathcal{P}_n$ for all large n, so $\limsup_n \left\|p(b^{(n)})\right\| = \limsup_n \left\|p(a^{(n)})\right\| = \|p(a)\|$ for every p. Therefore $a_i \mapsto [(b_i^{(n)})_n]$ is well defined and isometric on the $*$-algebra over $\mathbb{Q}(i)$ generated by the a_i, which is dense in A, and it extends to an isometric $*$-homomorphism of A into a norm matrix corona. □

Lemma 11.2 (tensor synchronization)**.** *Let Γ be a countable group regularly realized by (A_1, ρ_1, τ_1), let Q be a countable group with homomorphisms $\beta_n\colon Q \to B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large n, let $S \leq \Gamma$, and let $\tau\colon S \to Q$ be a homomorphism. Suppose that for every n there is a homomorphism $\lambda_n\colon \Gamma \to G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

Then there are a separable unital MF C^*-algebra A, a homomorphism $V\colon \Gamma \times Q \to \mathcal{U}(A)$, a tracial state T on A with $T(V(g,q)) = 0$ for $(g,q) \neq (1,1)$, and a unitary $W \in A$ with $WV(s,1)W^* = V(s,\tau(s))$ for all $s \in S$. Consequently

$$\langle \Gamma \times Q,\ u \ \mid\ u(s,1)u^{-1} = (s,\tau(s))\ (s \in S)\rangle$$

is regularly realized, and in particular MF.

Proof. Let E_n be the image of $(\lambda_n, \beta_n)\colon \Gamma \times Q \to G_n \times B_n$, a finite group of order k_n, and let L_n be its left regular representation on $\ell^2(E_n)$. Put $B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 11.1, let $E = \prod_n B'_n / \bigoplus_n B'_n$, and define

$$V(g,q) = \left[\left(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q))\right)_n\right], \qquad T([(x_n)_n]) = \lim_\omega (\tau_1 \otimes \mathrm{tr}_{k_n})(x_n),$$

for a free ultrafilter ω. Each coordinate of V is a homomorphism into a unitary group, so V is a homomorphism, and T is a tracial state on E because $|(\tau_1 \otimes \mathrm{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g,q)) = 0$; if

$g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large n, the regular representation of a finite group has trace zero off the identity, and again $T(V(1, q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n\tau(s))$ from S to E_n have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. Their images are therefore isomorphic subgroups of E_n of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n\tau(s))$. The restriction of L_n to a subgroup of E_n is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1)W_n^* = L_n(\lambda_n(s), \beta_n\tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s, 1)W^* = V(s, \tau(s))$ for $s \in S$. Let A be the C^*-algebra generated by $V(\Gamma \times Q)$ and W; it is separable and unital, and MF by Lemma 11.1. The triple $(A, V, T|_A)$ regularly realizes $\Gamma \times Q$. For the last assertion, apply Theorem 9.2 to this realization, with ι any injective $*$-homomorphism of A into a norm matrix corona and with the conjugator $\iota(W)$. □

Lemma 11.3 (the positive branch)**.** *If* $e \in \mathrm{INF}$*, then* $\widehat{R}_e$ *is regularly realized; in particular, it is MF.*

Proof. Let $e \in \mathrm{INF}$, so that $Q_e = H_0$, $N_e = N_+$, $q_e = q_+$, and $\tau_e(i(f)) = q_+(f)$, $\tau_e(v\, i(f)\, v^{-1}) = 1$. By Lemma 10.6 it suffices to show that R_e is regularly realized, and by Lemma 11.2 applied to $\Gamma = \Gamma_e$, $Q = H_0$, $S = S_e$ and $\tau = \tau_e$, it suffices to produce the data of that lemma.

The group K_e is residually finite by Lemma 10.5(1), hence regularly realized by Lemma 9.1, and Γ_e is the central HNN extension of K_e over L_e, hence regularly realized by Corollary 9.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \cdots$ of P with trivial intersection, let $r_n : P \to C_n = P/P_n$, and put $\beta_n = r_n \circ j : H_0 \to C_n$ with j the embedding of Lemma 10.3(3). Since j is injective and the P_n decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large n.

Let $G_n = (C_n \times C_n) \rtimes \langle\sigma\rangle$, where σ of order two exchanges the coordinates. Define λ_n on $K_e = K_e^0 \times (F \times P)$ by killing K_e^0 and sending (f, p) to $(r_n j q_+(f),\, r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (16): the factor L_e^0 of L_e lies in K_e^0 and is killed, and an element $(f, jq_+(f))$ of L^{g} is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with σ. Hence $\lambda_n : \Gamma_e \to G_n$ is a homomorphism, and on the generators of S_e,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \qquad \lambda_n(v\, i(f)\, v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1 : S_e \to H_0$ be the homomorphisms that are q_+ on one copy of F in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 10.5(3); thus $\pi_0 = \tau_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j\pi_0,\, r_n j\pi_1)$, both sides being homomorphisms that agree on generators. Therefore

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j\pi_0) \cap \ker(r_n j\pi_1) \leq \ker(\beta_n \circ \tau_e),$$

which is the hypothesis of Lemma 11.2. □

Proof of Theorem D. By Lemma 10.6 the map $e \mapsto \widehat{R}_e$ is computable, by Lemma 11.3 the presented group is MF for $e \in \mathrm{INF}$, and by Lemma 10.7 it is not MF for $e \in \mathrm{FIN}$. Hence INF reduces to $\mathrm{MF}_{\mathrm{fp}}$ and FIN reduces to $\mathrm{NONMF}_{\mathrm{fp}}$ under computable many-one reductions. Since INF is Π^0_2-complete and FIN is Σ^0_2-complete, and since $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ and $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$ by Proposition 8.2, the two sets are complete at their levels. A Π^0_2-complete set is not Σ^0_2 and a Σ^0_2-complete set is not Π^0_2, so neither set is recursively enumerable, and in particular neither is decidable. □

The proof of Lemma 11.3 never claims that the explicit unitaries $V(g, q)$ and W represent R_e faithfully, and indeed they do not: the finite maps λ_n kill the Mikhailova factor, so W commutes with $V(\gamma, 1)$ for every $\gamma \in K^0_e$, although u does not commute with $(\gamma, 1)$ in R_e. Faithfulness is restored at the universal algebra in Theorem 9.2. Three questions in the same register remain open. Whether the models of $\widehat{R}_e$ can be chosen to converge to $C^*_{\mathrm{r}}(\widehat{R}_e)$ in operator norm, so that the reduced C^*-algebra itself is MF, is not settled by the argument, which produces only an abstract regular trace. Whether $\widehat{R}_e$ is hyperlinear for $e \in \mathrm{INF}$ is open, and would place the recognition problem for hyperlinearity at the same level; by Theorem 9.2 and the argument above this is a question about the regular trace of the rope. Finally, whether the sofic analogue of the compiler exists is open, because the tensor synchronization has no counterpart for permutation models.

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Use of AI and formal methods

Claude Fable 5 and GPT-5.6 Sol were used extensively in mathematical exploration, proof development, manuscript drafting, literature navigation, Lean formalization, and editorial revision. The Lean development is available at `https://github.com/SauersML/group-approximation`; it builds with `lake build`, and its command `#audit_closed_axioms` prints the axioms on which any declaration depends, so that each endpoint named below can be checked to rest on the standard axioms alone.

For this paper, the development verifies the following.

- The one-sided transport theorem and the normal property-(T) radical criterion, which together give Theorem A.
- The rank 12 compression and commutator calculations and the resulting full MF radical conclusion of Theorem B.

- For Proposition 6.1, a direct proof that every nontrivial normal subgroup contains an elementary matrix $e_{ij}(x)$ with $x \neq 0$, using the concrete two-sided sandwich and the fact that every central unit of the binary Leavitt algebra over $\mathbb{F}_2$ equals 1; the article cites Preusser's theorem instead, and the development formalizes that theorem separately over the hypothesis that the coefficient ring admits finite right exchange partitions.
- The quotient calculus of Theorem C.
- For Theorem D, the certificate normal form and the passage from a computable finite-output compiler to Π^0_2-completeness; the compiler of Section 10 and Theorem 9.2 are proved in the article.
- A machine-readable manifest lists every numbered statement of this paper and records, for each, whether it is matched by a Lean declaration or proved in the article.

Beyond this paper, the development proves the following closed, kernel-audited statements.

- Soficity of a recursively enumerated presentation is Π^0_2-complete; the hardness half follows the construction of Bilanovic–Chubb–Roven [BCR] from a finitely presented nonsofic group, whose nonsoficity is that of the Leavitt unit group of OpenAI's chapter [OAI], re-proved in the development from Kun's expander decomposition and the Kun–Thom criterion.
- There is one fixed finitely presented non-MF group with a computable family of words w_p, each invisible to every operator norm matrix model and lying in the commutator of the MF radical with the group, such that $w_p = 1$ exactly when the program p halts.
- There is a finitely presented, point-norm matricially stable, non-MF group whose MF radical is the normal closure of a single nontrivial element.
- Property (T) for $\mathrm{SL}_3(\mathbb{Z})$, from an exact rational sum-of-squares certificate.
- The level-three congruence subgroup $\Gamma(3)$ of $\mathrm{SL}_3(\mathbb{Z})$ is torsion-free, finitely presented, and has property (T).
- Shalom's theorem that every finitely generated group with property (T) is a quotient of a finitely presented group with property (T).

The development is organized for reuse: the group properties are the predicates `IsOperatorMF`, `IsSofic` and `HasKazhdanPropertyT`, the statements of this paper are collected under `Manuscript/OneSidedMFRadical`, and new results can be stated against the same definitions and audited with the same command.

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